

EDUCATION

Institute for Interdisciplinary Information Science, Tsinghua University.	Beijing, China.
PhD(candidate) in Computer Science and Technology. Advisor: Andrew Chi-Chih Yao.	Sep 2021–present
Institute for Interdisciplinary Information Science, Tsinghua University.	Beijing, China.
Bachelor of Engineering in Computer Science and Technology, a.k.a. “Yao Class”.	Sep 2017–Jul 2021
GPA: 3.9/4.0. Thesis: <i>On the Random Walk Problem over Graphs</i> .	

PUBLICATIONS

1. Ran Duan, Jiayi Mao, Xiao Mao, Xinkai Shu, and **Longhui Yin**, “Breaking the sorting barrier for directed single-source shortest paths”, *57th Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, 2025.
Best Paper Award
2. Ran Duan, Jiayi Mao, Xinkai Shu, and **Longhui Yin**, “A randomized algorithm for single-source shortest path on undirected real-weighted graphs”, *IEEE 64th Annual Symposium on Foundations of Computer Science (FOCS)*, 2023.
3. Seth Pettie, Thatchaphol Saranurak, and **Longhui Yin**, “Optimal vertex connectivity oracles”, *54th Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, 2022.
4. Seth Pettie, Dingyu Wang, and **Longhui Yin**, “Non-mergeable sketching for cardinality estimation”, *48th International Colloquium on Automata, Languages, and Programming (ICALP)*, 2021.
5. Seth Pettie and **Longhui Yin**, “The structure of minimum vertex cuts”, *48th International Colloquium on Automata, Languages, and Programming, (ICALP)*, 2021.

EXPERIENCES

Citadel Securities Asia Services Limited.	Hong Kong, China.
Internship. Quantitative Research.	Jun 2024 - Sep 2024
Electrical Engineering and Computer Science Department, University of Michigan.	Ann Arbor, U.S.
Academic Visiting, advised by Professor Seth Pettie.	Feb 2020 - Aug 2020
On cardinality estimation and graph connectivity problems.	
Department of Information Engineering, Chinese University of Hong Kong.	Hong Kong, China.
Academic Visiting, advised by Professor Angela Yingjun Zhang.	Jul 2019 - Aug 2019
On approximate message passing and compressed sensing problems.	

SCHOLARSHIPS AND AWARDS

• Best Paper Award on 57th Annual ACM SIGACT Symposium on Theory of Computing (STOC)	2025
• Future Scholar Scholarship of Tsinghua University	2021
• Beijing Excellent Graduating Student Award	2021
• Yao Award – Bronze Prize	2020
• Tang Lixin Scholarship (only applicable for top 5% students)	2020
• National Encouragement Scholarship	2018 & 2019
• Tsinghua University Scholarship for Freshman	2017
• Gold Medal in Chinese Mathematical Olympiad(longlist of 60 for IMO 2017 Chinese Team)	2016